

What else is hindering the adoption of electric cars in India?

Different vehicles require different charging connectors, which is a significant hindrance. While the number of EV charge points is increasing in the country, finding one that supports all electric cars and has multiple connector types is rare.

Can standardisation of chargers help in the adoption of electric cars?

Smartphones have shown that a single type of charger pin for all is best for consumers. Consumers should not have to worry about whether electric charge points will support their specific electric car.

Technically, is it difficult to equip an EV charger with a universal connector to adapt to different power supplies independently?

Currently, yes, because different global regions have different protocols. Many connectors will not serve the purpose in such cases. There should be a single type of connector for all electric cars.

Many vehicle OEMs, including Mercedes Benz, have backtracked on their commitments regarding EV retail plans. Is there confusion in the enablers network for EVs as well?

Most of those backtracking are from the premium car segment. The mass market for cars in India falls between ₹800,000 and ₹1,200,000. Maruti Suzuki, the largest car seller in India, is currently working closely with many EV charger makers. Tata Motors, on the other hand, leads the electric car market in India. Combining the announcements made by vehicle OEMs for India, I don't see any confusion in the EV market. The situation will improve once

more mass-market electric cars are launched in the country.

Confusion around a product can create a lot of issues in its supply chain. How do you ensure that does not happen at Delta?

There are over 20 EV charger manufacturers operating in India today, and Delta is one of the few with in-house manufacturing of critical components. We manufacture over 80% of components for EV chargers in-house, including rectifiers and controller boards. Much of this manufacturing is also being shifted to India to comply with the Make In India initiative.

But some players say outsourcing components allows them to focus more on R&D and product advancements!

Picking up components from outside India and components with origins outside India makes it difficult for companies to adhere to compliances in India. Additionally, quality or performance issues arising in EV chargers due to a problem with one of the components will require one to run back to the component supplier. If the same is located outside India, it will mean unnecessary time spent on a process that could have been avoided altogether.

Do you see any advancements on the charger front to help consumers adopt more EVs?

Fast charging, home charging enablement, wireless charging, and battery swapping are anticipated future technologies. The industry is also working on chargers with multiple dispensing guns and portable fast chargers.

Where would you bet upon among these technologies?

For the immediate future, we

at Delta focus on chargers with multiple dispensing guns. I think this is the need of the hour, as a lot of use cases emerge from it. This addresses various use cases and can be a solution for societies with limited parking space, enabling charging of multiple vehicles simultaneously.

How would these work?

Let's suppose we have a one-megawatt power source that can be distributed in the 60kW, 120kW, or 360kW range. Based on power needs, we can distribute power in multiples of 60, 120, 360, and so on, accommodating mid- and high-range vehicles.

Will the dispensing guns house multiple types of connectors?

That can be done, but standardising connectors for all electric cars would simplify the process and ensure equal power distribution.

Is there a timeline for deployment?

We are currently conducting feasibility studies, with technology transfer completed. Safety aspects are being evaluated, and OEMs are showing significant interest, particularly in the fleet and electric bus sectors.

Do you foresee electric powertrain adoption in the truck segment?

There has been a lot of adoption in the EV truck segment (heavy duty and medium duty EV trucks segment). The leading brands have already started defining the launching dates.

Does Delta white-label for other brands?

We did when we started the EV charging venture under Delta, and we continue to do so for some charge point operators and vehicle OEMs. **EFY**